A company works with number of employees, all the works are dependents on the employees. Even if one of the employees resign the job immediately then assigned work will be not finished at the time, so delivery of the project to the clients will be delayed. Company planned to make solution for this, they want to know which employee may resign next. If they know previously, they can arrange alternative to avoid such problem. As an AI Engineer you must give Solution to this.

A) **How will you achieve this in AI?** First of all, will decide the problem statement that fall under which domain. Section B explains the way to figure out how to find.

Then using the sample dataset, which is present in the section D, we can see the common pattern that the employees who are joined recently and not get the hike and bonus much and utilize the annual and medical leaves than others leave the company. Using this can identify the potential employees who intent to leave soon.

B) **Find out the 3 -Stage of Problem Identification**:

1. Stage 01 = Since the dataset does not involve with images and only dealing with plan data we can go with Machine Learning.
2. Stage 02 = Please see the sample dataset in the section D, and the STATUS column has labelled clearly whether the employee is left or not. So, we go with Supervised Learning.
3. Stage 03 = The output column is NOT dealing with numbers (continuous numbers), instead it's using the ACTIVE/LEFT values (meaning some categorical data). So we can go with Classification.

C) **Name the project**: Employee-Churn-Application

D) **Create the dummy Dataset**

S/N	EMPLOYEE_NAME	JOINT_DATE	SALARY	LAST_HIKE	BONUS	PENDING ANNUAL LEAVES	PENDING MEDICAL LEAVES	STATUS
1	Karthic	1/2/2009	25,000.00	5.00%	5,000.00	11.00	14.00	ACTIVE
2	Danush	2/3/2008	41,000.00	6.50%	4,000.00	10.00	11.00	ACTIVE
3	Vijay	3/4/2007	38,000.00	7.00%	7,300.00	13.00	10.00	ACTIVE
4	Surya	4/5/2011	43,000.00	9.00%	12,000.00	9.00	12.00	ACTIVE
5	Silambarasan	5/6/2016	16,000.00	1.20%	1,000.00	3.00	2.00	LEFT
6	Prasanna	6/7/2019	12,000.00	0.50%	200.00	1.00	4.00	LEFT